ST 3011 - Statistical Programming
Individual Contribution Report - Group 8
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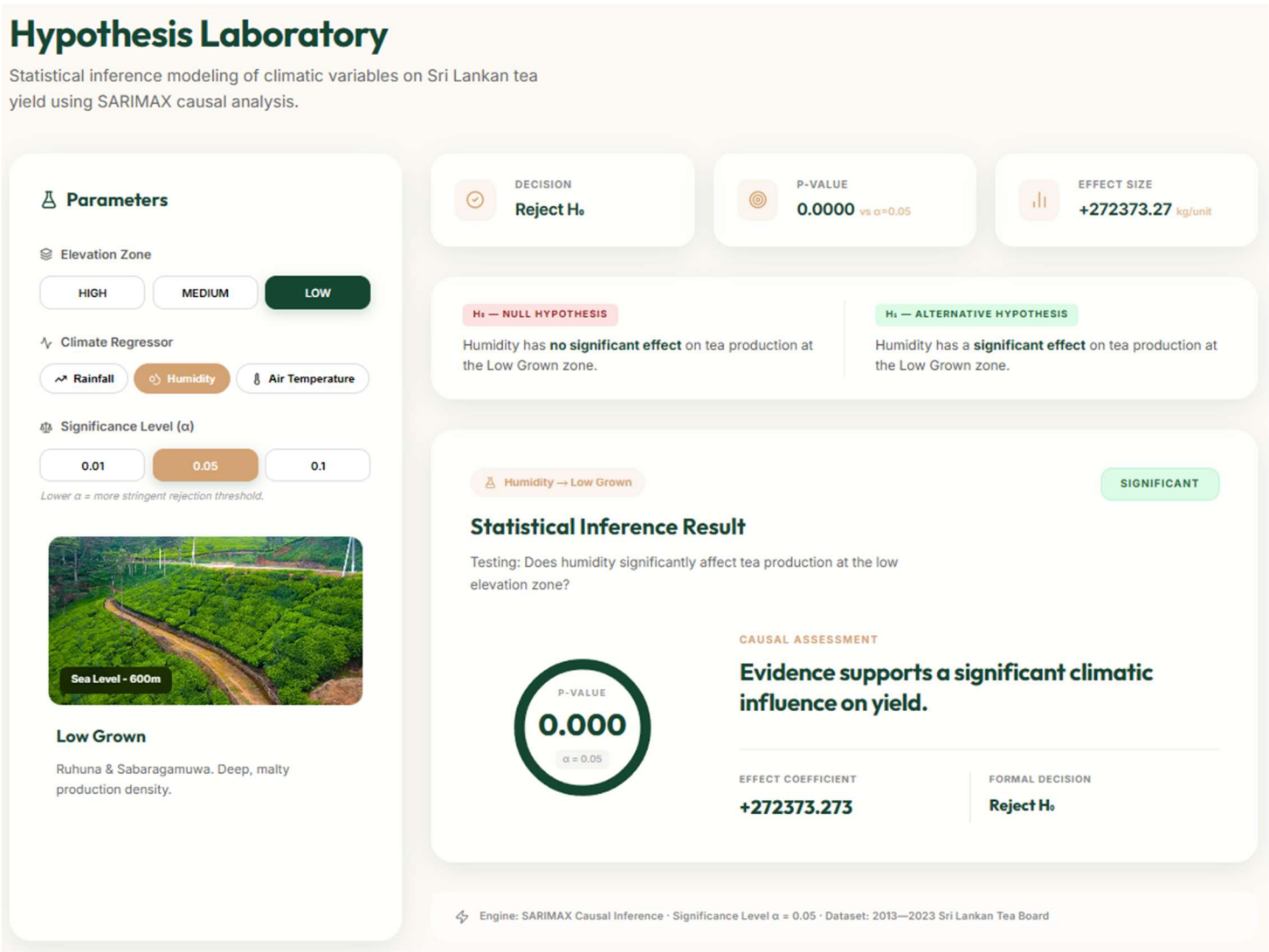
For the Tea Intelligence Platform, I led the development of the **Hypothesis Testing & Statistical Inference Module**, integrating both backend modeling and frontend visualization to deliver actionable insights on tea yield across different elevation zones.

On the **backend**, I implemented a **SARIMAX model** to estimate the impact of rainfall, temperature, and humidity. I also done key diagnostics, including **stationarity (ADF test)**, **residual autocorrelation (Ljung-Box test)**, and **multicollinearity (VIF)**, ensuring that coefficient estimates and p-values are statistically correct (This part is not visible in the website but it done for test the hypothesis assumptions). The system dynamically calculates results based on elevation category and user-selected significance levels, providing real time hypothesis testing outcomes.

On the **frontend**, I built a **glassmorphic dashboard** that allows users to toggle significance levels (α = 0.01, 0.05, 0.1) and instantly see whether hypotheses are rejected or not. Professional grade charts display coefficients , while smooth state management between Python and React ensures a responsive experience. A human friendly interpretation layer translates technical results into clear statements, e.g., *"Humidity significantly impacts low-elevation tea yield"*.